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DOOR LOCK DEVICE

Abstract

A door lock device includes a spring latch rotatably and movably disposed within a housing to be pressed to rotate and move between a projecting position, where the spring latch projects outwardly of a latch hole of the housing, and a retreated position, where the spring latch retreats inwardly into the latch hole. A retaining member is pivotably disposed to the housing, and is operable to rotate between a retaining position to engage with the spring latch to prevent rotation of the spring latch in the projecting position, and a releasing position to disengage from the spring latch. A biasing spring is disposed to bias the retaining member to the retaining position and to keep the retaining member in the retaining position.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Taiwanese Patent Application No. 113106817, filed on Feb. 26, 2024, and incorporated by reference herein in its entirety.

FIELD

[0002] The disclosure relates to a door lock device, and more particularly to a push-and-pull door lock device.

BACKGROUND

[0003] Conventional door locks have many components, thereby causing problems such as inconvenient fabrication, complicated construction and high manufacturing costs. Moreover, the deadbolt of a conventional door lock is prone to generate noises due to collision during the locking process, which results in discomfort and distress to users.

SUMMARY

[0004] Therefore, an object of the disclosure is to provide a door lock device that can alleviate at least one of the drawbacks of the prior art.

[0005] According to the disclosure, the door lock device includes a housing, a spring latch and a retaining biasing assembly. The housing defines a receiving space therein, and has a latch hole in communication with the receiving space. The spring latch is rotatably and movably disposed in the receiving space adjacent to the latch hole to be pressed to rotate and move between a projecting position, where a portion of the spring latch projects outwardly of the latch hole, and a retreated position, where the spring latch retreats inwardly into the latch hole. The retaining biasing assembly is disposed within the housing, and includes a retaining member which is pivotably disposed in the housing, and a biasing spring. The retaining member is operable to rotate between a retaining position, where the retaining member is engaged with the spring latch to prevent rotation of the spring latch in the projecting position, and a releasing position, where the retaining member is disengaged from the spring latch. The biasing spring is disposed to bias the retaining member to the retaining position and to keep the retaining member in the retaining position.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] Other features and advantages of the disclosure will become apparent in the following detailed description of the embodiment with reference to the accompanying drawings. It is noted that various features may not be drawn to scale.

[0007] FIG. 1 is a perspective view illustrating an embodiment of a door lock device according to the disclosure.

[0008] FIG. 2 is an exploded perspective view of the embodiment.

[0009] FIG. 3 is an exploded perspective view of the embodiment taken from another angle.

[0010] FIG. 4 is a fragmentary right side view of the embodiment.

[0011] FIG. 5 is a fragmentary left side view of the embodiment.

[0012] FIG. **6** is an exploded perspective view of a portion of the embodiment.
[0013] FIG. **7** is an exploded perspective view similar to FIG. **6**, but taken from another angle.
[0014] FIG. **8** is a fragmentary sectional view taken along line VIII-VIII of FIG. **1**.
[0015] FIG. **9** is a fragmentary sectional view taken along line IX-IX of FIG. **1**.
[0016] FIG. **10** is an exploded perspective view of a portion of the embodiment.
[0017] FIG. **11** is an exploded perspective view of a portion of the embodiment.
[0018] FIG. **12** is a fragmentary right side view illustrating the embodiment in a door-opened state.
[0019] FIG. **13** is a fragmentary sectional view similar to FIG. **8**, illustrating the embodiment in a door-closed state.
[0020] FIG. **14** is a fragmentary right side view illustrating a tongue in a retracted position.
[0021] FIG. **15** is a fragmentary right side view similar to FIG. **4**, illustrating a deadbolt assembly in an unlocking position.
[0022] FIG. **16** is a fragmentary right side view illustrating a buffering ring in a closed position.
[0023] FIG. **17** is a fragmentary right side view similar to FIG. **16**, illustrating the buffering ring in an opened position.

DETAILED DESCRIPTION

[0024] It should be noted herein that for clarity of description, spatially relative terms such as “top,” “bottom,” “upper,” “lower,” “on,” “above,” “over,” “downwardly,” “upwardly” and the like may be used throughout the disclosure while making reference to the features as illustrated in the drawings. The features may be oriented differently (e.g., rotated 90 degrees or at other orientations) and the spatially relative terms used herein may be interpreted accordingly. In the following description, a first horizontal direction (D1), a second horizontal direction (D2) and a vertical direction (D3) are transverse to one another. The first horizontal direction (D1) is substantially a front-rear direction, the second horizontal direction (D2) is substantially a left-right direction, and the vertical direction (D3) is substantially an up-down direction.

[0025] Referring to FIGS. **1** to **3**, an embodiment of a door lock device **100** according to the disclosure is of a push-and-pull door lock and is adapted to be mounted in a door plate (not shown). The door lock device **100** includes a housing **1**, a spring latch **2**, a retaining biasing assembly **3**, a speed-reduction mechanism **4**, a deadbolt assembly **5**, an electrically driving mechanism **6**, a key driving mechanism **7** and a lock tongue mechanism **8**.

[0026] The housing **1** includes a housing shell **11**, a cover plate **12**, a front plate **13**, a buffering ring **14** and a buffering ring **15**. The housing shell **11** has a right opening (**11a**). The cover plate **12** is coupled with the housing shell **11** to cover the right opening (**11a**). The front plate **13** is removably connected with front ends of the housing shell **11** and the cover plate **12** and cooperates with the housing shell **11** and the cover plate **12** to define an accommodation space **10** thereamong. The housing shell **11** has a base plate **110**. The base plate **110** has two elongated slots **111** spaced apart from each other in the first horizontal direction (D1) and each extending in the first horizontal direction (D1). The front plate **13** has an upper hole **131**, a plurality of deadbolt holes **132** formed below the upper hole **131** and spaced apart from each other in the vertical direction (D3), and a lower hole **133** formed below the deadbolt holes **132**. The buffering ring **14** defines a latch hole **141** extending therethrough, and is mounted on the front plate **13** to align the latch hole **141** with the upper hole **131**. The buffering ring **15** defines a tongue hole **151** extending therethrough, and is mounted on the front plate **13** to align the tongue hole **151** with the lower hole **133**. Each of the buffering rings **14**, **15** is made of an elastomeric material, such as rubber, silicone, etc.

[0027] With reference to FIGS. **4**, **5** and **6**, the housing shell **11** further has a support frame **112** disposed on the base plate **110** adjacent to the buffering ring **14**. The support frame **112** has a receiving space **113** therein in communication with the latch hole **141**, two elongated holes **114** in communication with a bottom of the receiving space **113** and spaced apart from each other in the second horizontal direction (D2), and a through hole **115** in communication with the receiving space **113** at an end opposite to the latch hole **141**. Each elongated hole **114** extends in the first

horizontal direction (D1). The housing shell **11** further has an axle shaft **116** disposed on the base plate **110** downwardly and rearwardly of the support frame **112**, and a threaded socket **117** disposed on the base plate **110** downwardly of the support frame **112**.

[0028] With reference to FIGS. **2**, **4**, **6** and **7**, the spring latch **2** is rotatably and movably disposed in the receiving space **113** adjacent to the latch hole **141** to be pressed to rotate and move between a projecting position (as shown in FIG. **4**), where a portion of the spring latch **2** projects outwardly of the latch hole **141**, and a retreated position (as shown in FIG. **13**), where the spring latch **2** retreats inwardly into the latch hole **141**. The spring latch **2** has a diamond-shaped cross-section, and is removably disposed within the housing **1** such that the spring latch **2** is selectively disposed in the receiving space **113** in one of a head-side direction and a tail-side direction which is opposite to the head-side direction by a 180-degree spin. Thus, the door lock device **100** may be applied to both the inward-opening door and the outward-opening door.

[0029] With reference to FIGS. **4**, **6**, **7** and **8**, the spring latch **2** includes a latch body **21**, a buffering pad **22** and an insert pin **23**. The latch body **21** has a diamond-shaped body **211** and at least two pairs of the protrusions **212**. The diamond-shaped body **211** has a triangular inner part **213** and a triangular outer part **214** extending from the inner part **213**. The inner part **213** is disposed in the receiving space **113** and has first and second side surfaces **215** opposite to each other in the vertical direction (D3). Three protrusions **212** project from the first side surface **215** and are spaced apart from one another in the second horizontal direction (D2) to cooperatively define two first grooves **216**, and three protrusions **212** project from the second side surface **215** and are spaced apart from one another in the second horizontal direction (D2) to cooperatively define two second grooves **216**. Hence, the first grooves **216** extend in the first horizontal direction (D1) and are spaced apart from each other in the second horizontal direction (D2), and the second grooves **216** extend in the first horizontal direction (D1) and are spaced apart from each other in the second horizontal direction (D2). The outer part **214** has a projecting end **210** projecting outwardly of the latch hole **141**, a recess **217** recessed from the projecting end **210**, and an insert hole **218** in communication with the recess **217** and extending in the vertical direction (D3). The buffering pad **22** is made of an elastomeric material, such as rubber, silicone, etc., and is engaged in the recess **217** and partly projects from the outer part **214** so as to contact a door frame (not shown). The buffering pad **22** has a through hole **221** which is aligned with the insert hole **218**. The insert pin **23** is engaged with and extends through the insert hole **218** and the through hole **221** to interfere with the outer part **214** and the buffering pad **22** so as to retain the buffering pad **22** on the latch body **21**.

[0030] With reference to FIGS. **4**, **5**, **6**, **7** and **8**, the retaining biasing assembly **3** is disposed in the accommodation space **10** of the housing **1**, and includes a retaining member **31** and a biasing spring **32**. The retaining member **31** is pivotably disposed on the axle shaft **116** of the housing **1**, and is operable to rotate between a retaining position (as shown in FIG. **4**), where the retaining member **31** is engaged with the spring latch **2** to prevent rotation of the spring latch **2** in the projecting position, and a releasing position (as shown in FIG. **12**), where the retaining member **31** is disengaged from the spring latch **2**. The biasing spring **32** is disposed to bias the retaining member **31** to the retaining position and to keep the retaining member **31** in the retaining position.

[0031] The retaining member **31** is an integrally formed single piece made by punching and bending machining processes. The retaining member **31** has a retaining shaft **311** disposed to retain the spring latch **2**, and a pressed shaft **312** extending from the retaining shaft **311**. The retaining shaft **311** has a U-shaped cross-section, and has two retaining ribs **313** spaced apart from each other in the second horizontal direction (D2) and each extending in the first horizontal direction (D1) to extend through the corresponding elongated hole **114** and to be engaged with the corresponding groove **216** of the spring latch **2**. The pressed shaft **312** has a transverse shaft section **314** extending rearwardly from the retaining shaft **311**, and an upright shaft section **315** extending transverse to and downwardly from the transverse shaft section **314**.

[0032] When the retaining member **31** is in the retaining position, the retaining ribs **313** are respectively engaged in the corresponding grooves **216** so as to restrict rotation of the spring latch **2** in the projecting position. When the retaining member **31** is in the releasing position, the retaining ribs **313** are disengaged from the corresponding grooves **216** to permit rotation of the spring latch **2** in the projecting position. With the first and second grooves **216** formed in the opposite first and second side surfaces **215**, the spring latch **2** may be disposed in the head-side direction so that the first grooves **216** may engage with the retaining ribs **313**, and the spring latch **2** may also be disposed in the tail-side direction so that the second grooves **216** may engage with the retaining ribs **313**. Therefore, the spring latch **2** may be disposed in a selected one of the head-side direction and the tail-side direction, and the retaining member **31** is disposed to retain the spring latch **2** in the projecting position.

[0033] The biasing spring **32** is a torsion spring which is sleeved around the threaded socket **117**, and has two ends abutting against the housing shell **11** and a bottom of the retaining shaft **311**.

[0034] With reference to FIGS. **4**, **5**, **6**, **7** and **9**, the speed-reduction mechanism **4** is disposed in the accommodation space **10** of the housing **1**, and includes a rack **41**, a sleeve **42**, a pinion **43**, a pinion shaft **44**, a bearing **45**, a first compression spring **46** and a second compression spring **47**. The rack **41** has a rack section **411**, an abutting section **412** and two sliding sections **413**. The rack section **411** extends in the first horizontal direction (D1) and through the through hole **115** of the support frame **112**. The abutting section **412** is connected with a front end of the rack section **411** and extends in the second horizontal direction (D2) to be disposed in the receiving space **113** of the support frame **112** and abut against the inner part **213** of the spring latch **2**. The sliding sections **413** extend from a left side of the rack section **411** and respectively and slidably engaged in the elongated slots **111** of the housing **1** (see FIG. **3**) so as to restrain the movement of the rack **41** only in the first horizontal direction (D1). The rack **41** is moved by spring latch **2** through the abutting engagement of the abutting section **412** with the inner part **213**.

[0035] The sleeve **42** is disposed on the base plate **110** of the housing shell **11** and rearwardly of the support frame **112**. The sleeve **42** has a sleeve wall **421**, two engaging slots **422** formed in the sleeve wall **421** spaced apart from each other in the first horizontal direction (D1), and an abutting surface **423** formed in and transverse to the sleeve wall **421** adjacent to the left side of the sleeve wall **421**. The pinion **43** and the pinion shaft **44** are received within the sleeve wall **421**. The pinion **43** is securely sleeved on the pinion shaft **44** and meshes with the rack section **411** of the rack **41**. The bearing **45** has a bearing body **451** received within the sleeve wall **421** to bear the pinion shaft **44**, and two engaging ribs **452** extending from the bearing body **451** and respectively engaged with the engaging slots **422** of the sleeve **42** so as to prevent rotation of the bearing **45** relative to the sleeve **42**.

[0036] The first compression spring **46** is received in the receiving space **113** of the support frame **112**, and has two ends in abutting engagement with the abutting section **412** of the rack **41** and the support frame **112** to apply a biasing force to urge the spring latch **2** toward the projecting position and to keep the spring latch **2** in the projecting position.

[0037] The second compression spring **47** is disposed within the sleeve wall **421** of the sleeve **42**, and has two ends abutting against the cover plate **12** and the bearing **45** to urge the bearing **45** to be in abutting engagement with the abutting surface **423** and to urge and keep the pinion **43** in mesh engagement with the rack section **411** of the rack **41**. Thus, disengagement of the pinion **43** from the rack section **411** during swaying of the bearing **45** and the pinion **43** in the sleeve **42** is prevented.

[0038] With reference to FIGS. **4**, **5** and **10**, the housing shell **11** further has a beam **118** which is disposed on the base plate **110** below the pressed shaft **312** of the retaining member **31**. The deadbolt assembly **5** is operably disposed in the accommodation space **10** of the housing **1**, and includes a sliding plate **51** slidably connected with the housing **1**, and a plurality of deadbolts **52** disposed on a front end of the sliding plate **51** and projecting from the housing **1**. The deadbolts **52**

are arranged in the vertical direction (D3), each projecting from the corresponding deadbolt hole **132**. The deadbolt assembly **5** is operably moved between a locking position (as shown in FIG. **4**), where a portion of each deadbolt **52** projects outwardly of the corresponding deadbolt hole **132**, and an unlocking position (as shown in FIG. **12**), where each deadbolt **52** retreats in the accommodation space **10** of the housing **1**. The sliding plate **51** has a notch **512** which extends in a direction of movement of the deadbolt assembly **5**, and first and second notch side surfaces **513**, **514** which are opposite to each other in the direction of the movement to cooperatively define the notch **512**. The first notch side surface **513** is interposed between the deadbolt **52** and the second notch side surface **514**.

[0039] The housing shell **11** further has an upper pivot axle **119** disposed on the base plate **110** above the beam **118**, and a lower pivot axle **120** disposed on the base plate **110** below and rearwardly of the beam **118**. The electrically driving mechanism **6** is disposed in the accommodation space **10** of the housing **1**, and includes an electrically driving assembly **61**, a transmitting pinion **62** and a cam member **63**. Referring to FIG. **5**, the electrically driving assembly **61** has a circuit assembly **611**, an electric motor **612** connected with the circuit assembly **611**, and a gear assembly **613** coupled with the electric motor **612**. The transmitting pinion **62** is pivotably connected with the upper pivot axle **119** and meshes with the gear assembly **613** to be driven by the electrically driving assembly **61** to rotate. Referring to FIGS. **4** and **10**, the transmitting pinion **62** has a toothed portion **621** which is pivotably connected with the upper pivot axle **119** and meshes with the gear assembly **613**, a first pressing protrusion **622** which projects from a right side of the toothed portion **621**, and a second pressing protrusion **623** which projects from the right side of the toothed portion **621** and which is angularly spaced apart from the first pressing protrusion **622**. The first pressing protrusion **622** is interposed between the transverse shaft section **314** of the pressed shaft **312** and the second pressing protrusion **623** to press the transverse shaft section **314** to rotate the retaining member **31** from the retaining position to the releasing position.

[0040] The cam member **63** has a pivot ring **631** which is pivotably connected with the upper pivot axle **119**, and a press arm **632** which extends radially and outwardly from the pivot ring **631**. The first pressing protrusion **622** is interposed between the pressed shaft **312** and the press arm **632**. The press arm **632** is interposed between the first pressing protrusion **622** and the second pressing protrusion **623**, and is engaged in the notch **512** of the sliding plate **51** to press the first notch side surface **513** of the sliding plate **51** to move the deadbolt assembly **5** toward the locking position, and to press the second notch side surface **514** to move the deadbolt assembly **5** toward the unlocking position. The cam member **63** is swingable, through driving of the transmitting pinion **62**, between a first swung position (as shown in FIG. **4**), where the deadbolt assembly **5** is in the locking position, and a second swung position (as shown in FIG. **12**), where the deadbolt assembly **5** is in the unlocking position.

[0041] Furthermore, the first pressing protrusion **622** of the transmitting pinion **62** is disposed to press the press arm **632** to swing the cam member **63** from the second swung position to the first swung position. The second pressing protrusion **623** is disposed to press the press arm **632** to swing the cam member **63** from the first swung position to the second swung position.

[0042] The key driving mechanism **7** is disposed within the housing **1**, and includes a rotary member **71** and a transmitting plate **72**. The rotary member **71** is pivotably connected with the base plate **110** and the cover plate **12**, and has a rotary cylinder **711** formed with a key hole **713**, and a driving arm **712** extending radially and outwardly from the rotary cylinder **711**. A key (not shown) is inserted into the key hole **713** and is operated to rotate the rotary member **71**.

[0043] The transmitting plate **72** is pivotably connected with the lower pivot axle **120** and is rotated with the rotary member **71**. The transmitting plate **72** has a plate body **721** which is pivotably connected with the lower pivot axle **120**, and a pressing protuberance **722** which projects from a side surface of the plate body **721** adjacent to a top end of the plate body **721**. The plate body **721** has a pressing edge **723** rearwardly of the pressing protuberance **722**, and an arcuate slot **724** for

the beam **118** to slidably extend therethrough. The pressing protuberance **722** is formed to press the upright shaft section **315** of the pressed shaft **312** so as to rotate the retaining member **31** from the retaining position to the releasing position. The pressing edge **723** is formed to press the press arm **632** to swing the cam member **63** from the first swung position to the second swung position. [0044] Specifically, with the rotation of the rotary member **71** by a key, the transmitting plate **72** is rotated between an initial position (as shown in FIG. 4), where the pressing edge **723** and the pressing protuberance **722** are respectively disengaged from the press arm **632** and the upright shaft section **315**, and an abutting position (as shown in FIG. 15), where the pressing edge **723** is in abutting engagement with the press arm **632** to retain the cam member **63** in the second swung position, and where the pressing protuberance **722** is in abutting engagement with the upright shaft section **315** to retain the retaining member **31** in the releasing position.

[0045] With reference to FIGS. 2, 4 and 5, the lock tongue mechanism **8** is disposed in the accommodation space **10** of the housing **1** and partly projects from the housing **1**. The lock tongue mechanism **8** includes a mounting plate **81**, a sliding rod **82**, a tongue **83**, a compression spring **84** and a sensing unit **85**. The mounting plate **81** is disposed on the base plate **110** of the housing shell **11**. The sliding rod **82** is slidably mounted on the mounting plate **81**. The tongue **83** is pivotably connected with the sliding rod **82** and partly projects from the tongue hole **151**. The compression spring **84** is sleeved around the sliding rod **82** and abuts against the sliding rod **82** and the mounting plate **81**. The sensing unit **85** includes a circuit board **851** which is electronically connected with the circuit assembly **611** of the electrically driving assembly **61** in a wired or wireless manner, and a switch **852** which is disposed on the circuit board **851** to be in contact with the sliding rod **82**. The tongue **83** is operable to move between an extended position (as shown in FIG. 4), where the tongue **83** partly projects from the housing **1** and the sliding rod **82** contacts the switch **852**, and a retracted position (as shown in FIG. 14), where the tongue **83** retracts into the housing **1** and the sliding rod **82** is disengaged from the switch **852**. The switch **852** generates a signal as the sliding rod **82** is disengaged from the switch **852**, which is sent by the circuit board **851** to the circuit assembly **611** of the electrically driving assembly **61** to control the electrically driving assembly **61** to rotate the transmitting pinion **62**. The transmitting pinion **62** brings the cam member **63** to swing to thereby press the deadbolt assembly **5** to the locking position.

[0046] With reference to FIGS. 11, specifically, the tongue **83** includes a tongue body **831**, a buffering pad **832** and an insert pin **833**. The tongue body **831** has a substantially triangular cross-section, and is pivotably connected with the sliding rod **82**. The tongue body **831** has an extended end **830** which projects outwardly of the tongue hole **151**, a recess **834** which is recessed from the extended end **830**, and an insert hole **835** which is in communication with the recess **834**. The buffering pad **832** is made of an elastomeric material, such as rubber, silicone, etc., and is engaged in the recess **834** and partly projects from the tongue body **831** so as to contact the door frame (not shown). The buffering pad **832** has a through hole **836** which is aligned with the insert hole **835** in the vertical direction (D3). The insert pin **833** is engaged with and extends through the insert hole **835** and the through hole **836** to interfere with the tongue body **831** and the buffering pad **832** to retain the buffering pad **832** on the tongue body **831**.

[0047] With reference to FIG. 12, when the door panel is in a door-opened state, the first pressing protrusion **622** of the transmitting pinion **62** abuts and presses a lower end of the transverse shaft section **314** of the retaining member **31** to retain the retaining member **31** in the releasing position. At this stage, the biasing spring **32** is compressed and deformed to create a biasing returning force. The second pressing protrusion **623** abuts and presses a front end of the press arm **632** to retain the cam member **63** in the second swung position. At this stage, the press arm **632** abuts and presses the second notch side surface **514** of the sliding plate **51** to retain the deadbolt assembly **5** in the unlocking position. Moreover, the spring latch **2** is kept in the projecting position, and the tongue **83** is kept in the extended position.

[0048] With reference to FIGS. 9, 12, 13 and 14, during rotation of the door panel to the door frame

to the door-closed state, the buffering pad **22** of the spring latch **2** contacts the door frame, the spring latch **2** is rotated by the door frame and moved inwardly to the retreated position. During the rotation and movement of the spring latch **2**, the rack **41** is slid inwardly to bring the pinion **43** to rotate, and the first compression spring **46** is compressed by the abutting section **412** of the rack **41** with a biasing returning force. With the frictional engagement of the pinion **43** with the rack section **411** and through controlling a friction force between the pinion shaft **44** and the bearing **45**, and the first compression spring **46** having two ends abutting against the abutting section **412** and the support frame **112**, the sliding movement of the rack **41** may be buffered and a speed of the sliding movement may be reduced so as to reduce noises generated during the retreated movement of the spring latch **2** and the rack **41**.

[0049] Meanwhile, the tongue **83** contacts the door frame, and is pressed and rotated by the door frame to retract to the retracted position. The sliding rod **82** is slid inwardly during the rotation and movement of the tongue **83**, and the compression spring **84** is compressed to create a biasing returning force. The sliding rod **82** is disengaged from the switch **852** such that the switch **852** generates a signal, which is sent by the circuit board **851** to the circuit assembly **611** of the electrically driving assembly **61**.

[0050] With reference to FIGS. **4**, **5** and **8**, when the door panel is in the door-closed state, the rack **41** and the spring latch **2** are biased by the biasing returning force of the first compression spring **46** such that the spring latch **2** is rotated and moved outwardly to automatically return to the projecting position. At this stage, the spring latch **2** is retained in the door frame. Moreover, the circuit assembly **611** controls the operation of the electric motor **612** to rotate the transmitting pinion **62** in a first rotational direction (R1) through the gear assembly **613**. During the rotation of the transmitting pinion **62**, the first pressing protrusion **622** is moved away from the transverse shaft section **314**, and the retaining member **31** is biased by the biasing returning force of the biasing spring **32** to automatically rotate about the axle shaft **116** in the first rotational direction (R1) to the retaining position. With the engagement of the retaining ribs **313** of the retaining member **31** with the corresponding first or second grooves **216**, the spring latch **2** is retained in the projecting position to prevent rotation thereof.

[0051] Subsequently, the first pressing protrusion **622** abuts against and presses the press arm **632** of the cam member **63** to swing the cam member **63** in the first rotational direction (R1) from the second swung position. During swinging of the cam member **63**, the press arm **632** presses the first notch side surface **513** of the sliding plate **51** to move the deadbolt assembly **5** in the first horizontal direction (D1) to the locking position. The rotation of the transmitting pinion **62** is stopped when the cam member **63** is swung to the first swung position, such that the cam member **63** is retained in the first swung position while the deadbolt assembly **5** is retained in the locking position where the deadbolts **52** are retained in the door frame. Thus, the door panel is locked to the door frame through the door lock device **100** (as shown in FIG. **1**).

[0052] With reference to FIGS. **5** and **12**, an unlocking operation of the door lock device **100** through the electrically driving mechanism **6** is described as follows. The circuit assembly **611** of the electrically driving assembly **61**, when receiving an unlocking signal, controls the operation of the electric motor **612** to rotate the transmitting pinion **62** in a second rotational direction (R2) opposite to the first rotational direction (R1) through the gear assembly **613**. During the rotation of the transmitting pinion **62**, the first pressing protrusion **622** abuts against and presses the transverse shaft section **314** to rotate the retaining member **31** about the axle shaft **116** in the second rotational direction (R2), and the second pressing protrusion **623** abuts against and presses the press arm **632** to swing the cam member **63** in the second rotational direction (R2) from the first swung position. During the swinging of the cam member **63**, the press arm **632** presses the second notch side surface **514** of the sliding plate **51** to move the deadbolt assembly **5** in the first horizontal direction (D1) toward the unlocking position. The rotation of the transmitting pinion **62** is stopped when the cam member **63** is swung to the second swung position, such that the retaining member **31** is

retained in the releasing position, and the cam member **63** is retained in the second swung position while the deadbolt assembly **5** is retained in the unlocking position where the deadbolts **52** are moved away from the door frame.

[0053] With reference to FIGS. **1** and **15**, an unlocking operation of the door lock device **100** through the key driving mechanism **7** is described as follows. The rotary member **71** is rotated in the first rotational direction (R1) through a key driving action. During the rotation of the rotary member **71**, the driving arm **712** presses the transmitting plate **72** to rotate about the lower pivot axle **120** in the first rotational direction (R1) while the pressing protuberance **722** presses the upright shaft section **315** of the retaining member **31** to rotate the retaining member **31** about the axle shaft **116** in the second rotational direction (R2), and the pressing edge **723** presses the press arm **632** to swing the cam member **63** in the second rotational direction (R2). During the swing the cam member **63**, the press arm **632** presses the second notch side surface **514** of the sliding plate **51** to move the deadbolt assembly **5** in the first horizontal direction (D1) toward the unlocking position. When the transmitting plate **72** is rotated to the abutting position, the retaining member **31** is retained in the releasing position, the cam member **63** is retained in the second swung position while the deadbolt assembly **5** is retained in the unlocking position where the deadbolts **52** are moved away from the door frame.

[0054] With reference to FIGS. **12**, **13** and **15**, in the above unlocking state of the door lock device **100**, the door panel may be pushed away from the door frame. The buffering pad **22** of the spring latch **2** contacts and is pressed by the door frame, which brings the spring latch **2** to rotate and move inwardly to the retreated position. Once the door panel is opened, the spring latch **2** is urged by the biasing returning force of the first compression spring **46** to automatically move back to the projecting position. Additionally, the sliding rod **82** and the tongue **83** are urged by the biasing returning force of the compression spring **84** to bring the tongue **83** back to the extended position and the sliding rod **82** to contact the switch **852**.

[0055] With reference to FIGS. **1**, **4** and **6**, during opening and closing of the door panel, with the buffering pad **22** and the buffering pad **832** contacting the door frame, the latch body **21** and the tongue body **831** may be prevented from direct collision with the door frame and generating noises so as to achieve a silent working effect. Moreover, with the buffering ring **14** stopping the protrusions **212** of the spring latch **2**, the spring tongue **2** is prevented from disengagement from the latch hole **141** to retain the spring latch **2** in the projecting position, thereby achieving buffering and muffling effects in operation. Furthermore, with the buffering ring **15** stopping the tongue body **831** of the tongue **83** during rotation and movement of the tongue **83**, the tongue **83** may be prevented from disengagement from the tongue hole **151** to retain the tongue **83** in the extended position, thereby achieving buffering and muffling effects.

[0056] With reference to FIGS. **2**, **3**, **16** and **17**, the base plate **110** of the housing shell **11** has a first axial hole (**110a**) below the support frame **112**. The cover plate **12** has a second axial hole **121** aligned with the first axial hole (**110a**). The buffering ring **14** includes a ring body **142** which defines the latch hole **141**, and a pivot axle **143** which is connected with a lower end of the ring body **142**. The ring body **142** is disposed in the upper hole **131** of the front plate **13** to stop the protrusions **212**. The pivot axle **143** is pivotably engaged in the first axle hole (**110a**) and the second axial hole **121**. Thus, when the front plate **13** is removed from the housing shell **11** and the cover plate **12**, the buffering ring **14** may be rotated relative to the housing shell **11** and the cover plate **12** between a closed position (as shown in FIG. **16**) and an opened position (as shown in FIG. **17**). In the closed position, the latch hole **141** is aligned with the upper hole **131** for projection of the diamond-shaped body **211** therethrough, and the ring body **142** is disposed to stop the protrusions **212**. In the opened position, the latch hole **141** is moved away from and disengaged from the diamond-shaped body **211**, and the buffering ring **14** is moved away and disengaged from the protrusions **212**.

[0057] When it is desired to change the direction of the door lock device **100** so that the door lock

device **100** may be applied to either one of the inward-opening door or the outward-opening door, the front plate **13** is firstly removed from the housing shell **11** and the cover plate **12** to disengage the buffering ring **14** from the upper hole **131**. The spring latch **2** is then pressed inwardly to the retreated position to disengage the protrusions **212** from the ring body **142**. Subsequently, the buffering ring **14** is pulled outwardly and is rotated to the opened position so as to disengage the latch hole **141** from the diamond-shaped body **211**.

[0058] Next, the spring latch **2** is removed from the receiving space **113**, is turned by 180 degrees, and is inserted into the receiving space **113**. Then, the buffering ring **14** is pulled inwardly and rotated back to the closed position. Finally, the front plate **13** is mounted back on the housing shell **11** and the cover plate **12**. With the spring latch **2** having a diamond-shaped cross-section, the spring latch **2** is mountable within the housing **1** in either the head-side direction or the tail-side direction, and is rotatable in the latch hole **141** and the receiving space **113**. The spring latch **2** may be rotated and moved in the latch hole **141** and the receiving space **113** without any pivoting mechanism. Therefore, the structure of the spring latch **2** and the housing **1** is simple in reducing the manufacturing costs. Moreover, the direction changing operation of the spring latch **2** is simple and convenient to conduct.

[0059] In various embodiments, two pairs of the protrusions **212** are formed on the first and second side surfaces **215**. Specifically, the protrusions **212** of each pair project from the corresponding one of the first and second side surfaces **215** and cooperatively define the corresponding one of the first and second grooves **216**. The retaining shaft **311** has only one retaining rib **313**.

[0060] Alternatively, the sleeve **42** has one engaging slot **422**, and the bearing **45** has one engaging rib **452**.

[0061] As illustrated, since the housing **1**, the spring latch **2** and the retaining member **3** are configured and arranged in the above structural relationship, the door lock device **100** has a fewer number of components and is convenient to fabricate, and is simple and compact in structure in reducing the manufacturing costs. With the spring latch **2** being selectively mountable in the receiving space **113** in one of a head-side direction and a tail-side direction which is opposite to the head-side direction by a 180-degree spin, the door lock device **100** may be applied to both the inward-opening door and the outward-opening door. Through the buffering pads **22**, **832** contacting the door frame, the latch body **21** and the tongue body **831** may be prevented from direct collision with the door frame and generating noises so as to achieve a silent working effect. With the buffering ring **14** stopping the protrusions **212** of the spring latch **2**, the spring tongue **2** is prevented from disengagement from the latch hole **141**, thereby achieving buffering and muffling effects in operation. Furthermore, with the buffering ring **15** stopping the tongue body **831** of the tongue **83**, the tongue **83** may be prevented from disengagement from the tongue hole **151**, thereby achieving buffering and muffling effects. With the speed-reduction mechanism **4**, noises generated during the retreating movement of the spring latch **2** is reduced.

[0062] While the disclosure has been described in connection with what is considered the exemplary embodiment, it is understood that this disclosure is not limited to the disclosed embodiment but is intended to cover various arrangements included within the spirit and scope of the broadest interpretation so as to encompass all such modifications and equivalent arrangements.

Claims

1. A door lock device, comprising: a housing defining a receiving space therein, and having a latch hole in communication with said receiving space; a spring latch rotatably and movably disposed in said receiving space adjacent to said latch hole to be pressed to rotate and move between a projecting position, where a portion of said spring latch projects outwardly of said latch hole, and a retreated position, where said spring latch retreats inwardly into said latch hole; and a retaining biasing assembly disposed within said housing, said retaining biasing assembly including a

retaining member which is pivotably disposed in said housing, and a biasing spring, said retaining member being operable to rotate between a retaining position, where said retaining member is engaged with said spring latch to prevent rotation of said spring latch in the projecting position, and a releasing position, where said retaining member is disengaged from said spring latch, said biasing spring being disposed to bias said retaining member to the retaining position and to keep said retaining member in the retaining position.

2. The door lock device of claim 1, wherein said spring latch has a diamond-shaped cross-section, and is removably disposed within said housing such that said spring latch is selectively disposed in said receiving space in one of a head-side direction and a tail-side direction which is opposite to the head-side direction by a 180-degree spin.

3. The door lock device of claim 2, wherein said spring latch has a first groove and a second groove opposite to each other, said retaining member being engaged with said first groove when said spring latch is disposed in the head-side direction, and being engaged with said second groove when said spring latch is disposed in the tail-side direction.

4. The door lock device of claim 3, wherein said housing includes a buffering ring, said buffering ring defining said latch hole extending therethrough, said spring latch having a diamond-shaped body and at least two pairs of said protrusions, said diamond-shaped body being disposed in said receiving space and extendable through said latch hole, said diamond-shaped body having two side surfaces opposite to each other, said protrusions of each pair projecting from a corresponding one of said side surfaces and cooperatively defining a corresponding one of said first groove and said second groove, said buffering ring being disposed to stop said protrusions during rotation and movement of said spring latch to retain said spring latch in the projecting position.

5. The door lock device of claim 4, wherein said buffering ring is rotatable between a closed position, where said latch hole is disposed for projection of said diamond-shaped body therethrough, and said buffering ring is disposed to stop said protrusions, and an opened position, where said latch hole is moved away and disengaged from said diamond-shaped body, and said buffering ring is moved away and disengaged from said protrusions.

6. The door lock device of claim 5, wherein said housing includes a housing shell having a right opening, a cover plate coupled with said housing shell to cover said right opening, and a front plate removably connected with front ends of said housing shell and said cover plate, said housing shell having a first axial hole, said cover plate having a second axial hole, said front plate having an upper hole, said buffering ring including a ring body which defines said latch hole and which is mounted on said front plate to align said latch hole with said upper hole and to stop said protrusions, and a pivot axle which is connected with said ring body and which is pivotably disposed in said first axial hole and said second axial hole such that, when said front plate is removed from said housing shell and said cover plate, said buffering ring is rotatable relative to said housing shell and said cover plate between the closed position and the opened position.

7. The door lock device of claim 1, wherein said spring latch includes a latch body and a buffering pad which is disposed on said latch body, said latch body including an outer part which has a projecting end projecting outwardly of said latch hole, a recess recessed from said projecting end, and an insert hole in communication with said recess, said buffering pad being engaged in said recess and partly projecting from said outer part, said buffering pad having a through hole which is aligned with said insert hole, said spring latch further including an insert pin which is engaged with and extends through said insert hole and said through hole to retain said buffering pad on said latch body.

8. The door lock device of claim 1, further comprising a speed-reduction mechanism disposed within said housing, said speed-reduction mechanism including a rack, a pinion, a pinion shaft and a bearing, said rack being slidably disposed on said housing and having an end in abutting engagement with said spring latch to be moved with said spring latch, said pinion being securely sleeved on said pinion shaft and meshing with said rack, said bearing being disposed to bear said

pinion shaft.

9. The door lock device of claim 8, wherein said speed-reduction mechanism further includes a first compression spring having an end in abutting engagement with said rack to apply a biasing force to urge said spring latch toward the projecting position.

10. The door lock device of claim 8, wherein said speed-reduction mechanism further includes a sleeve and a second compression spring, said sleeve being disposed on said housing for receiving said bearing therein, and having an abutting surface which is disposed therein, said second compression spring being disposed in said sleeve and abutting against said housing and said bearing to urge said bearing to be in abutting engagement with said abutting surface and urge said pinion in mesh engagement with said rack, said sleeve having at least one engaging slot, said bearing having at least one engaging rib which is engaged with said engaging slot to retain said bearing on said sleeve.

11. The door lock device of claim 1, further comprising a deadbolt assembly and an electrically driving mechanism, said deadbolt assembly being operably disposed within said housing to be operably moved between a locking position, where a portion of said deadbolt assembly projects outwardly of said housing, and an unlocking position, where said deadbolt assembly retreats in said housing, said electrically driving mechanism including an electrically driving assembly, a transmitting pinion which is pivotably connected with said housing and which is driven by said electrically driving assembly to rotate, and a cam member which is pivotably connected with said housing and which is swingable through said transmitting pinion, said transmitting pinion being rotated to press said retaining member to rotate from the retaining position to the releasing position, said cam member being swung to press said deadbolt assembly to move between the locking position and the unlocking position.

12. The door lock device of claim 11, wherein said retaining member is an integrally formed single piece, said retaining member having a retaining shaft disposed to retain said spring latch, and a pressed shaft extending from said retaining shaft, said cam member having a press arm which presses said deadbolt assembly such that said cam member being swingable between a first swung position, where said deadbolt assembly is in the locking position, and a second swung position, where said deadbolt assembly is in the unlocking position, said transmitting pinion having a toothed portion which is coupled with and driven by said electrically driving assembly, a first pressing protrusion which projects from a side of said toothed portion and which is interposed between said pressed shaft and said press arm, and a second pressing protrusion which projects from said side of said toothed portion and which is angularly spaced apart from said first pressing protrusion, said press arm being interposed between said first pressing protrusion and said second pressing protrusion, said first pressing protrusion pressing said pressed shaft to rotate said retaining member from the retaining position to the releasing position, and also pressing said press arm to swing said cam member from the second swung position to the first swung position, said second pressing protrusion pressing said press arm to swing said cam member from the first swung position to the second swung position.

13. The door lock device of claim 12, wherein said deadbolt assembly includes a sliding plate slidably connected with said housing, and at least one deadbolt disposed on said sliding plate and projecting from said housing, said sliding plate having a notch which extends in a direction of movement of said deadbolt assembly, and a first notch side surface and a second notch side surface which are opposite to each other in the in direction of the movement to cooperatively define said notch, said first notch side surface being interposed between said deadbolt and said second notch side surface, said press arm being engaged in said notch to press said first notch side surface to move said deadbolt assembly toward the locking position, and to press said second notch side surface to move said deadbolt assembly toward the unlocking position.

14. The door lock device of claim 12, wherein said pressed shaft includes a transverse shaft section to be pressed by said first pressing protrusion, and an upright shaft section extending transverse to

said transverse shaft section, said door lock device further comprising a key driving mechanism disposed within said housing, said key driving mechanism including a rotary member which is pivotably connected with said housing, and a transmitting plate which is pivotably connected with said housing and which is rotated with said rotary member to press said upright shaft section to rotate said retaining member from the retaining position to the releasing position, said transmitting plate also pressing said press arm to swing said cam member from the first swung position to the second swung position.

15. The door lock device of claim 14, wherein said transmitting plate has a plate body which is pivotably connected with said housing, and a pressing protuberance which projects from a side surface of said plate body and which is formed to press said upright shaft section, said plate body having a pressing edge which is formed to press said press arm, said transmitting plate being rotated with said rotary member between an initial position, where said pressing edge and said pressing protuberance are respectively disengaged from said press arm and said upright shaft section, and an abutting position, where said pressing edge is in abutting engagement with said press arm to retain said cam member in the second swung position, and where said pressing protuberance is in abutting engagement with said upright shaft section to retain said retaining member in the releasing position.

16. The door lock device of claim 1, further comprising a key driving mechanism disposed within said housing, said key driving mechanism including a rotary member which is pivotably connected with said housing, and a transmitting plate which is pivotably connected with said housing and which is rotated with said rotary member to press and rotate said retaining member from the retaining position to the releasing position, said retaining member having a retaining shaft disposed to retain said spring latch, and a pressed shaft extending from said retaining shaft to be pressed by said transmitting plate.

17. The door lock device of claim 11, further comprising a lock tongue mechanism which is disposed within said housing and partly projecting from said housing, said lock tongue mechanism being electronically connected with said electrically driving assembly such that said lock tongue mechanism generates a signal as a result of a contact changing action, which is sent to and controls said electrically driving assembly to bring rotation of said transmitting pinion so as to swing said cam member and press said deadbolt assembly to the locking position.

18. The door lock device of claim 17, wherein said lock tongue mechanism includes a mounting plate disposed on said housing, a sliding rod slidably mounted on said mounting plate, a tongue pivotably connected with said sliding rod and partly projecting from said housing, a compression spring sleeved around said sliding rod and abutting against said sliding rod and said mounting plate, and a sensing unit, said sensing unit including a circuit board which is electronically connected with said electrically driving assembly, and a switch which is disposed on said circuit board to be in contact with said sliding rod, wherein said tongue is operable to move between an extended position, where said tongue partly projects from said housing and said sliding rod contacts said switch, and a retracted position, where said tongue retracts into said housing and said sliding rod is disengaged from said switch, and wherein said switch generates the signal as said sliding rod is disengaged from said switch, which is sent by said circuit board to said electrically driving assembly.

19. The door lock device of claim 18, wherein said housing includes a buffering ring, said buffering ring defining a tongue hole extending therethrough, said buffering ring being disposed to stop said tongue during rotation and movement of said tongue to retain said tongue in the extended position.

20. The door lock device of claim 18, wherein said tongue includes a tongue body and a buffering pad which is disposed on said tongue body, said tongue body having an extended end projecting outwardly of said tongue hole, a recess recessed from said extended end, and an insert hole in communication with said recess, said buffering pad being engaged in said recess and partly projecting from said tongue body, said buffering pad having a through hole which is aligned with

said insert hole, said tongue further including an insert pin which is engaged with and extends through said insert hole and said through hole to retain said buffering pad on said tongue body.
